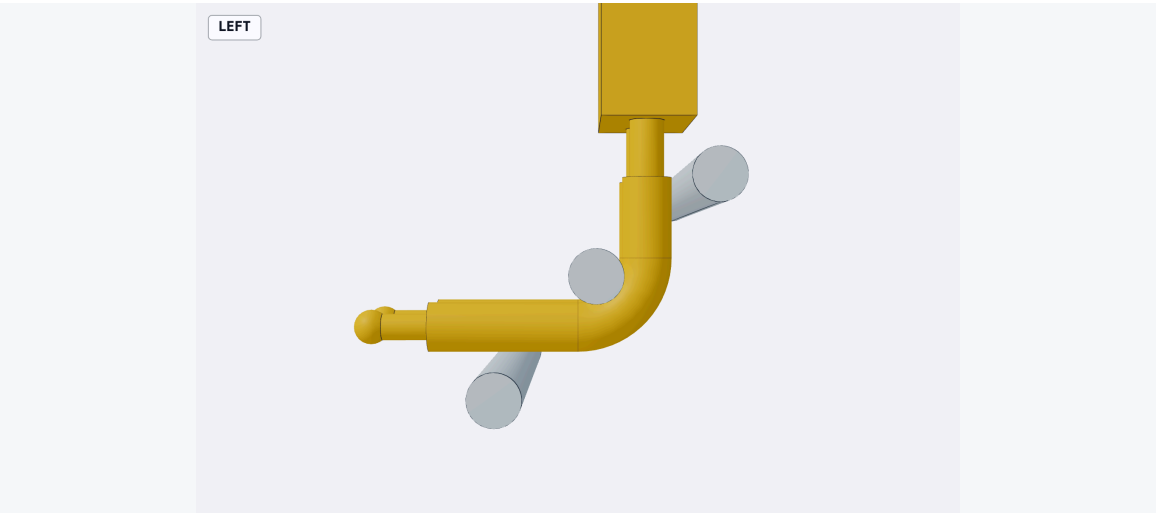
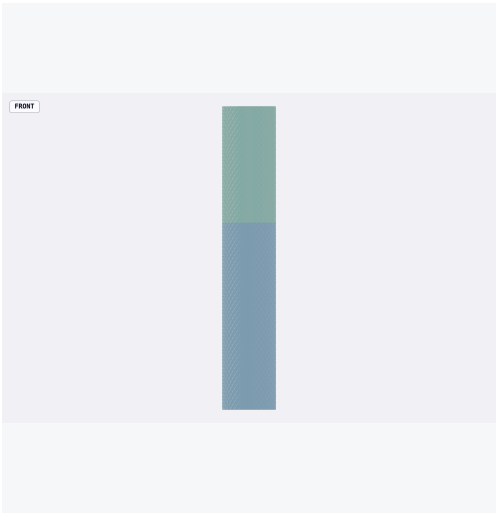
My 3D Printing Journey With AI Agents

Making physical things, and learning how to work together.



XT60 wire-bending jig · CAD



Laifen toothbrush case · R6 CAD

I wanted to make physical things

An idea
in my head

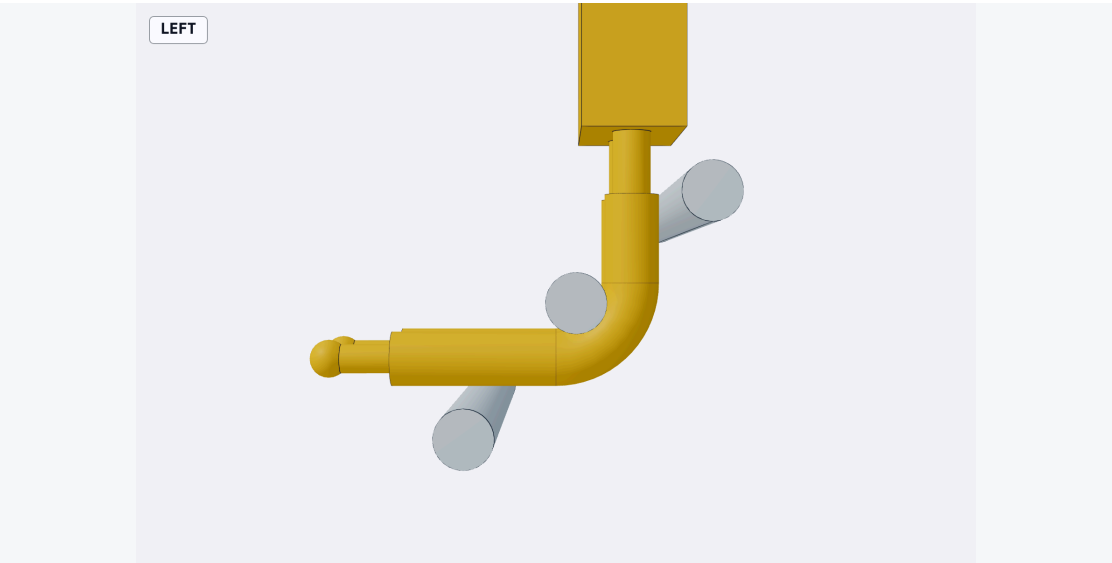
→ CAD
→ Orientation & supports
→ Transfer & printing
→ Materials
→ Slicing & settings
→ Monitoring

An object
I can use

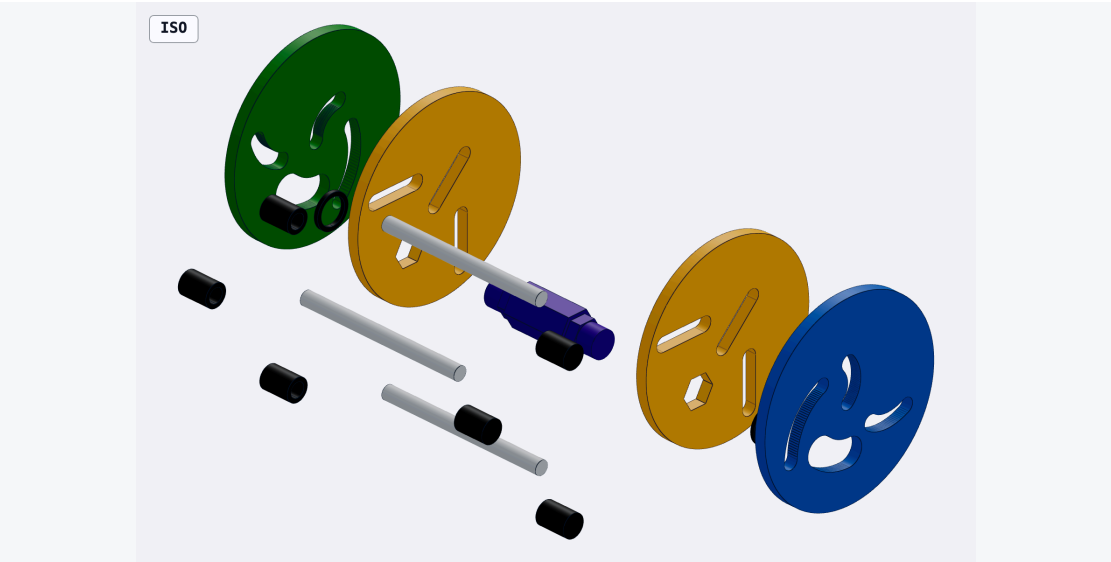
I did not set out to master every tool in between.

Those skills still matter. I want to spend more attention on **intent, constraints, taste and judgment**.

A jig for bending two wires



P13 contact cutaway · gold wire reference, grey forming rods

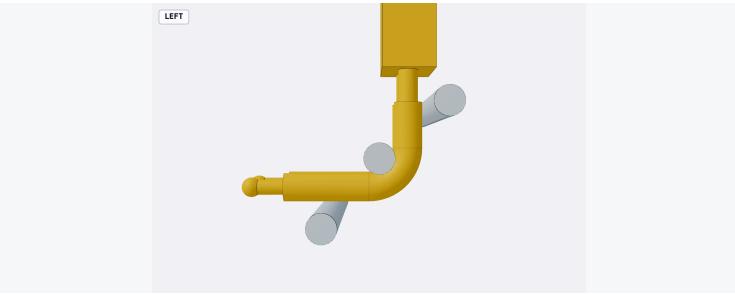


P18 exploded CAD · plates and grooves guide the rods

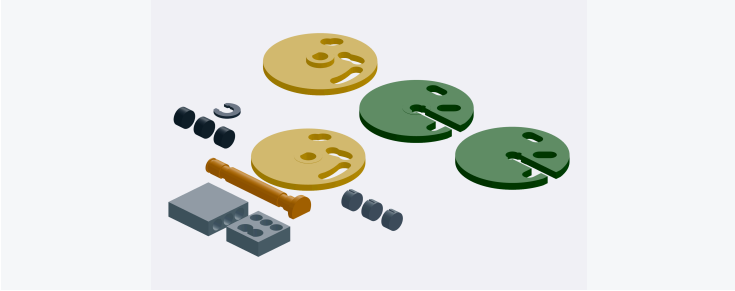
Hold two insulated wires in a **90° bend** while soldering and cooling.

XT60 is the connector on the two wires. Rods shape them; rails—grooves in the plates—guide movement.

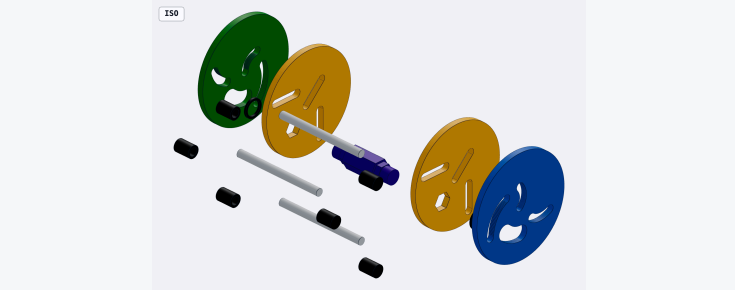
The design kept changing



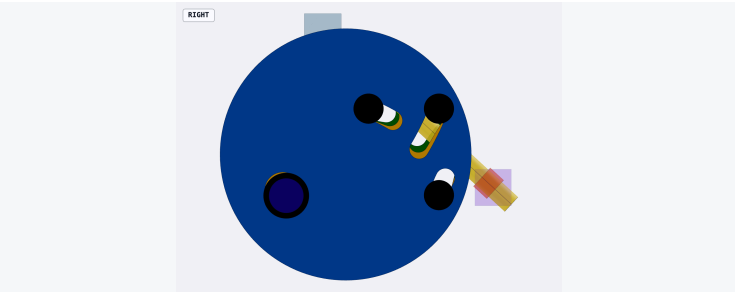
P13 · wire contact · CAD



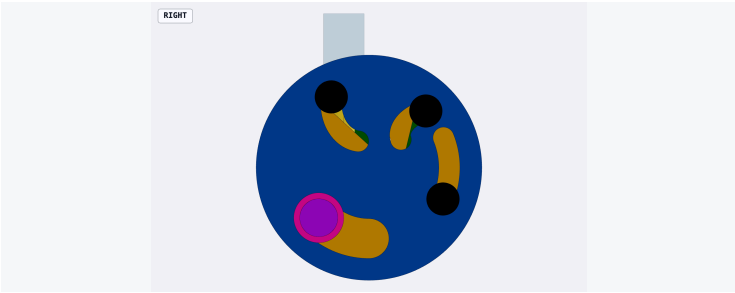
P13 · printable kit · CAD



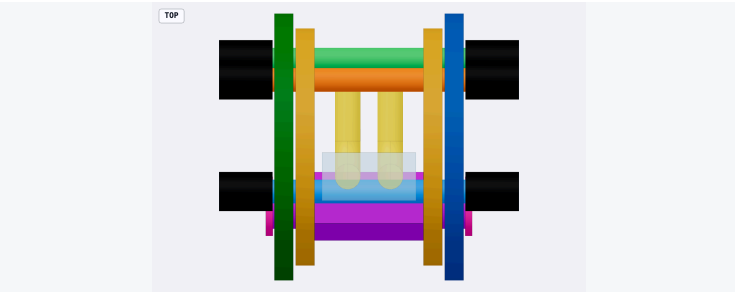
P18 · plate arrangement · CAD



P19 · shorter rails · CAD



P20 · revised curves · CAD

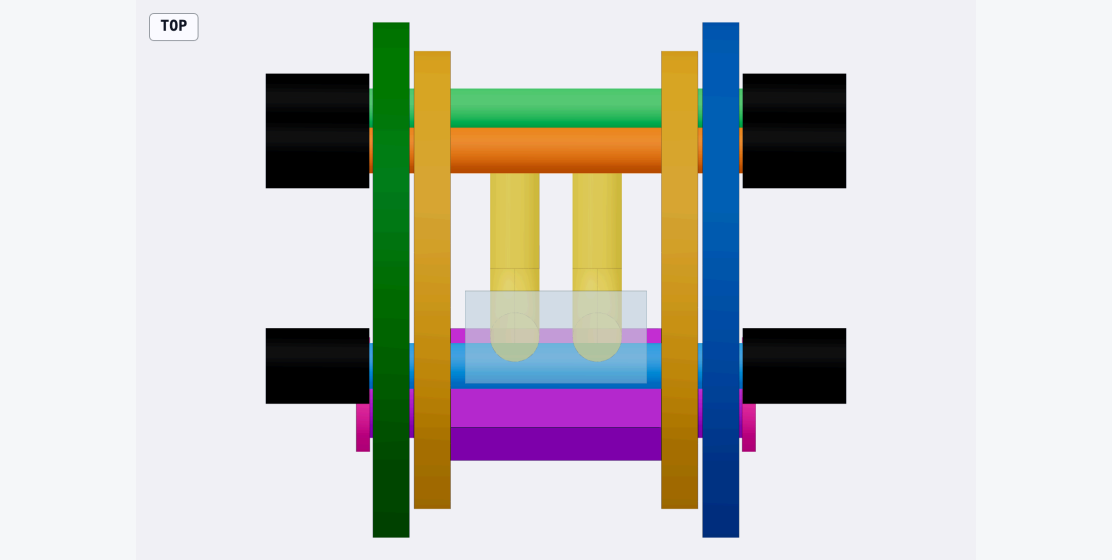


P20 · axial stack · CAD

25 recorded iteration folders in the audited project snapshot.

Folders include refinements, experiments and migration work. This is not a count of prints.

The real jig changed the problem



Earlier P20 axial CAD · before the width correction

More room	The reported inner span needed another 10 mm.
Different fit	Caps were too tight. Purchased rods had to stay 50 mm.
A different way to hold	Pushing felt smooth; releasing it sprang the mechanism open.

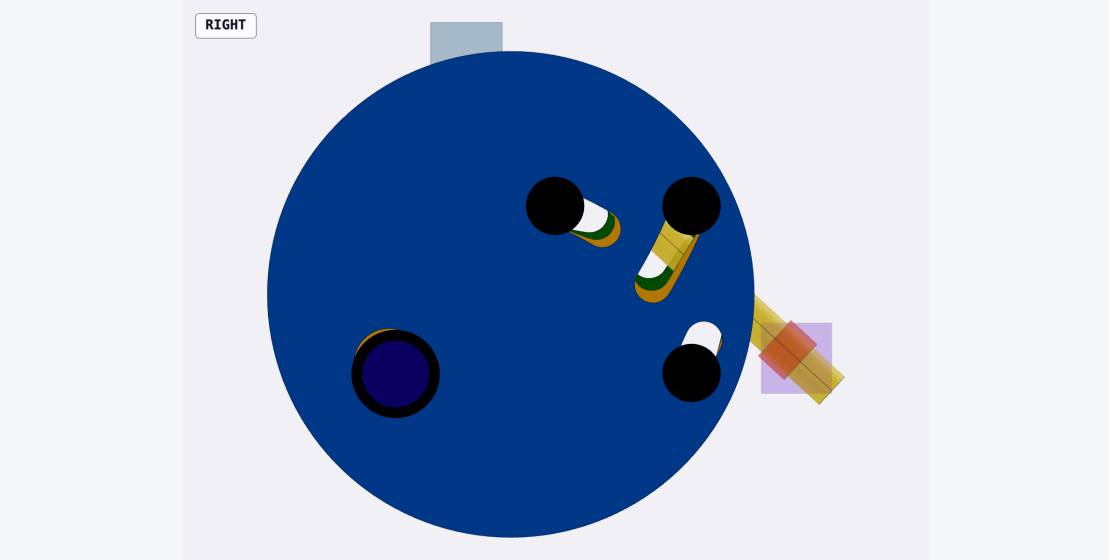
The next design let fingers position the rods, then used a clamp to hold them.

Recorded physical feedback, Aug28; later P24 intent preview still needed physical testing.

I kept asking for another rail route

I relied on AI to deliver the mechanism I imagined.

After many frustrating rail iterations, I still had not checked whether the relationships made sense.



One historical rail design · P19 CAD

An expensive route when the uncertainty stays vague · illustrative

Describe the idea → Generate a model → Print & assemble → Explain what went wrong → Try again

Simulation made the relationships visible

Me · intent and observations ↘ same representation ↙ **Agent** · explanation and revision



Actual later XT60 interactive study · P25 clamped state

This later study shows the kind of shared interface. It is not the missing original rail customizer.

Rear cradle

Supports the Blue and Green rods in this pose.

Front pad

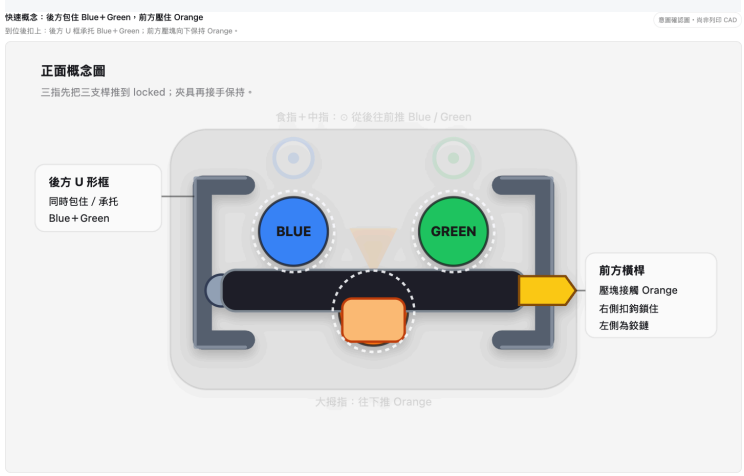
Holds the Orange rod.

We could point to **which part touches which**.

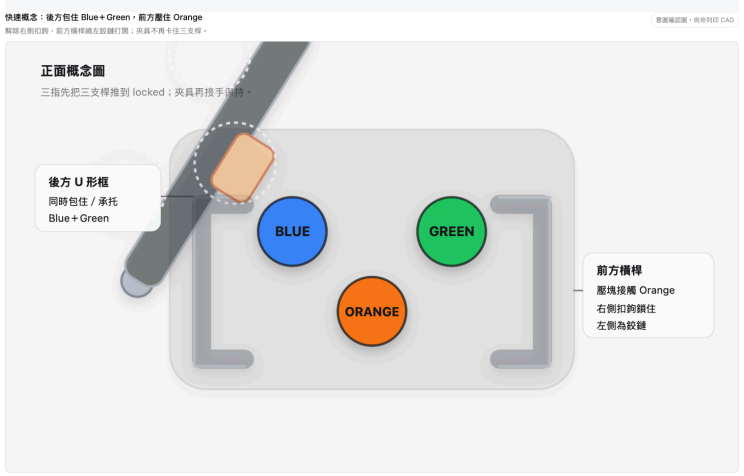
We could inspect the whole sequence



1 · Position the rods with fingers



2 · Bring the cradle and pad into contact

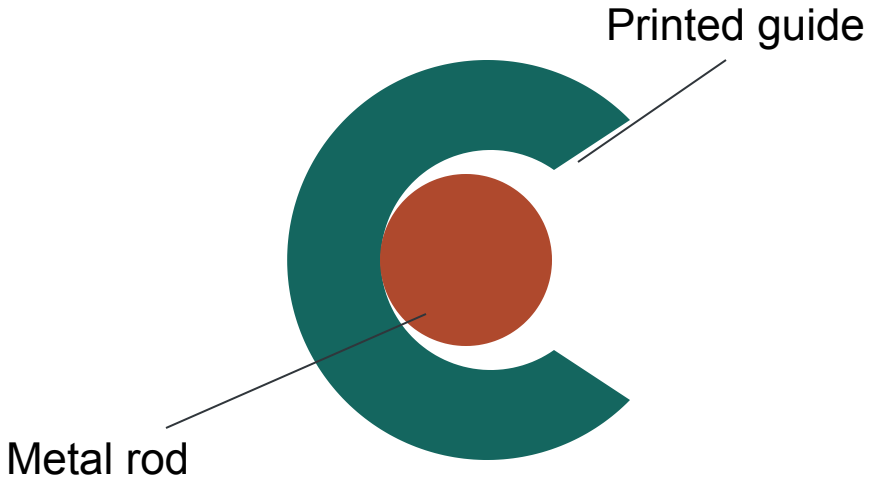


3 · Open the front and move the rear away

Sometimes the next thing to build is **the feedback interface**.

Actual P25 study states. Prescribed poses clarify intent; they do not calculate force, friction, fit or holding strength.

A small print answered one fit question



Read-only 3D cutaway · gap exaggerated ×8.
Teaching geometry; printed fit comes from the test.

A real 4 mm metal rod

Five model holes, printed in PETG with a 0.4 mm nozzle.

Model hole	Recorded hand fit
4.05 mm	Very tight
4.10 mm	Push / pull, still tight
4.15 mm	Best sliding; slight friction
4.20 mm	Looser
4.25 mm	Loose; wobble

A useful sliding-fit reference came from **the rod in the print**.

Observed coupon results, Aug21. Not a universal tolerance or whole-jig acceptance; no coupon photograph was recovered.

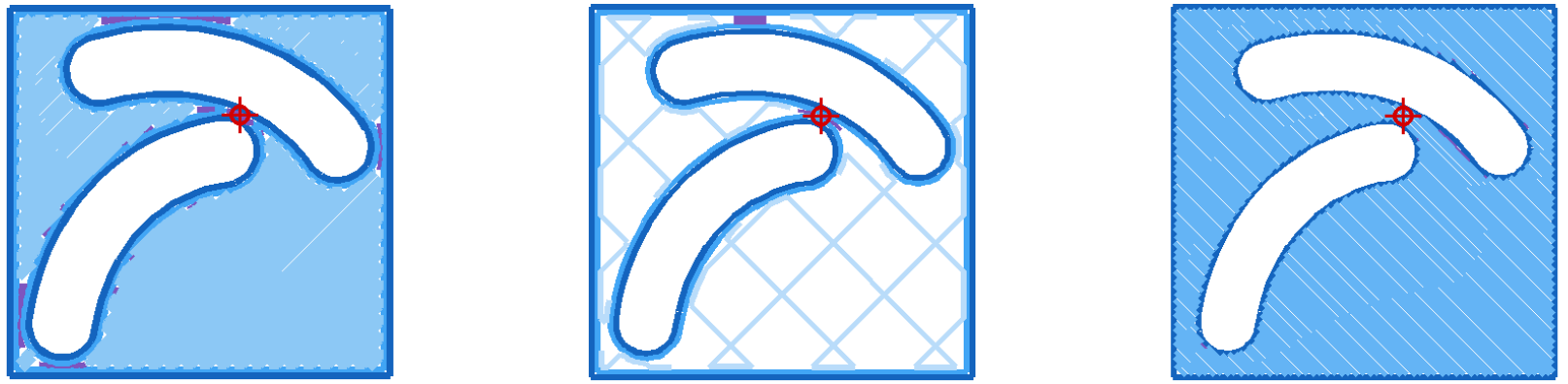
Another experiment isolated a thin web

Keep the real curved slots. Test whether the small bridge between them survives printing and handling.

FIRST layer 1
Z=0.25 mm; witness XY=(138.450, 141.703)

MIDDLE layer 8
Z=1.65 mm; witness XY=(138.450, 141.703)

TOP layer 16
Z=3.25 mm; witness XY=(138.450, 141.703)



External locator only - red cross is the 1.715 mm Blue-Orange minimum web; not part of coupon geometry.

Actual prepared toolpaths · first, middle and top layers

6m12s
slicer estimate

1.80g
slicer estimate

Physical result pending

Different questions need different evidence

Evidence	A useful question	Relative effort / speed	Still unknown
Reasoning / calculation	Are the numbers consistent?	Lowest / immediate	Real behavior
Simulation	Do the relationships make sense?	Low / fast to vary	Unmodelled physics
CAD inspection	Do these solids interfere?	Low–medium / digital	Printed fit
Partial print	Does this interface work?	Material + waiting	Whole assembly
Full print	Does it assemble and operate?	More material + waiting	Long-term use
Real use	Does it work for me?	Context + time	Other conditions

Choose the smallest experiment that can change the decision.

Illustrative relative costs; effort depends on the question. Digital tests do not stand in for physical evidence.

The useful loop starts with a specific gap

Observed + desired → Name the gap → State a hypothesis → Choose needed evidence

Digital experiment

Calculation · prototype · CAD inspection

↓ **model evidence**

Physical experiment

Partial print · assembly · use

↓ **physical observations**

Person + agent interpret the evidence

Update the shared model → choose the next action ↻

A focused loop changes what we ask for

Expensive and lossy

Broad request

↓ Whole model

↓ Full print & assembly

↓ Vague failure report

↓ Another whole revision

The unresolved relationship survives the loop.

Focused and shared

One uncertainty

↓ Shared diagram or small sample

↓ Specific observation

↓ Person + agent interpret

↓ Bounded next decision

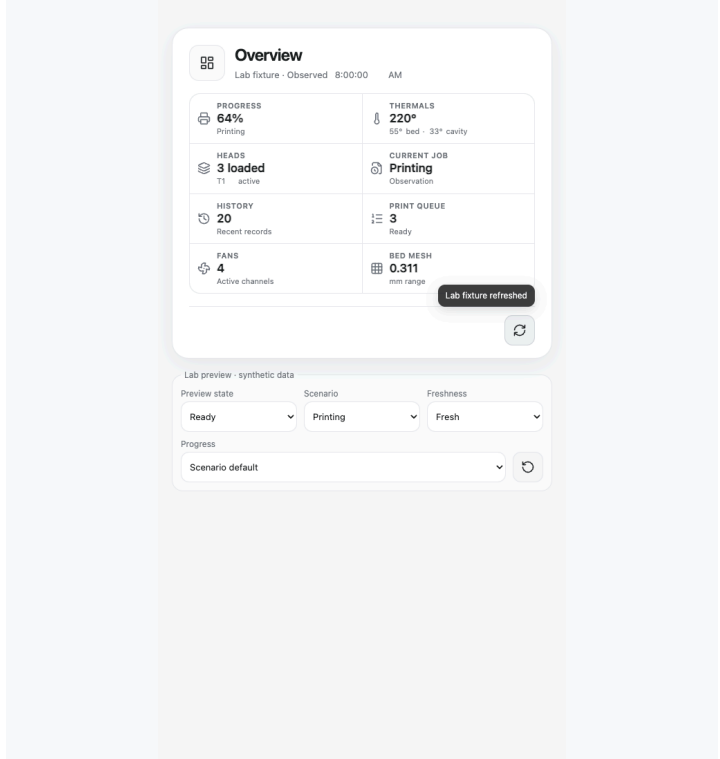
The evidence answers the question we chose.

Illustrative comparison. Effort and feedback strength depend on the experiment, not simply its size.

The execution tools can sit beneath the agent

Layer	What the evidence supports
Snapmaker Orca CLI guided workflows	Slice → inspect → authorized upload/start → monitor
U1 Print Studio MCP	Read-only observation of printer status and views
Further automation	Reusable preparation, support/orientation choices and reliable Live remain incomplete or unproven

The guided workflows have operated a real slicer and printer. The MCP app currently helps us observe.



U1 Print Studio UI · synthetic test fixture

Recorded runs and implementation audit, Sep3 snapshot. Fixture readings are not current printer telemetry.

A finished print still needs a person

One recorded PETG pilot: two cam-plate parts.

Execution result	Physical result
The print completed	Assembly exposed fit and mechanism-design failures

Agents can remove me from the toolchain **without removing me from the loop.**

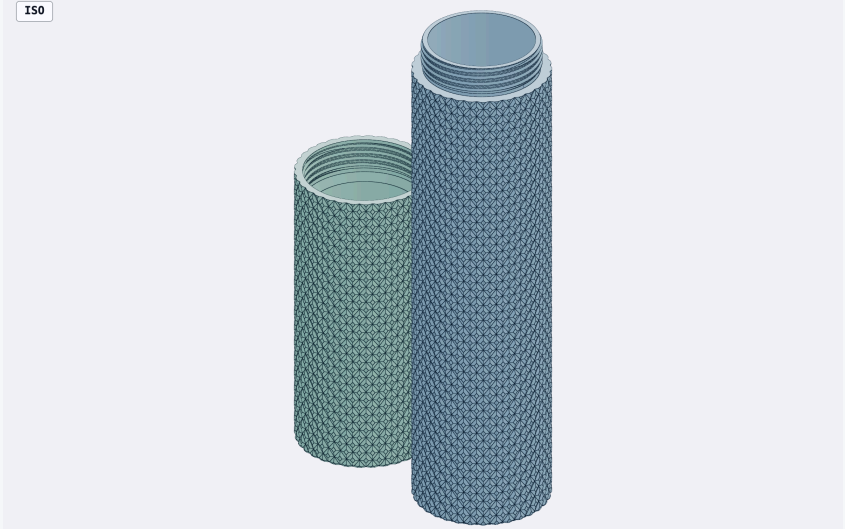
I still decide what matters, interpret fit and use, and accept the result.

Laifen: start with the parts I can hold

A travel case for my Laifen electric toothbrush.



Earlier physical thread specimens · actual photo

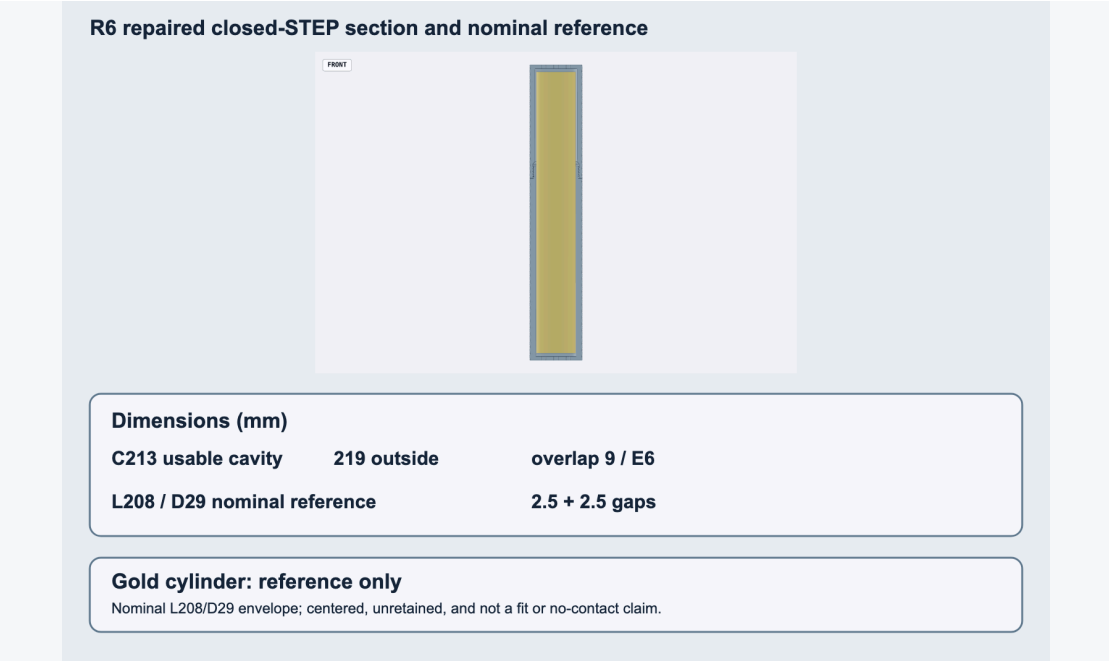


R6 two-part full case · reviewed CAD

All three printed closures felt good by hand. I wanted a uniform shape and a full-size case.

Earlier closure feedback does not establish physical fit of this R6 case.

Extra length does not hold the brush in place

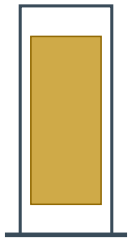


R6 section · nominal brush envelope, centered and unretained

Like XT60, this is a relationship we can inspect before more execution.

Centered

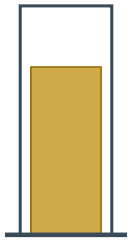
2.5 mm above



2.5 mm below

Resting on bottom

5 mm above



0 mm below · contact

Nominal schematic · gaps exaggerated

Earlier browser study · not R6 CAD controls

The next decision: is bottom support acceptable for the real loaded brush.

The same approach works for software

Illustrative experiment · not a claim this test has been completed

Uncertainty

A stale sensor reading may look current.

→

Small fixture

Feed a prototype an old timestamp and a failed refresh.

→

Evidence

Inspect the age label and failure state on screen.

If stale data is clearly marked:
test the next failure case.

If it still looks live:
revise the display before integration.

Plan the evidence before building the whole feature.

I want to choose what to learn next

One physical uncertainty

Does the jig keep holding after I release my hand.



Future load-sensor experiment

Record holding force during release.

A useful next decision

Compare the reading and observed motion with the holding criterion we agreed.

A camera can show motion. A sensor can add a measurement. An arm could repeat an action.

Future examples, not deployed integrations. People still choose the experiment and interpret what it means.

Keep human judgment in the feedback loop

Person
Intent · constraints · taste · acceptance

↔

Agent
Reasoning · tool use · continuity

Shared model of the problem

↓ **choose what to learn next**

Digital experiment
Calculate · simulate · inspect
↓ model evidence

Physical experiment + tools
Prepare · print · measure · use
↓ physical observations

↑ Interpret together and update the shared model ↻

Design feedback from day one.
Choose the fastest, cheapest, strongest useful loop.